

Tumelo Konaite | Software Engineer

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Location: South Africa, Johannesburg | [LinkedIn](#) | [GitHub](#) | [Website](#)

Summary

Software Engineer focused on building production-ready backend systems, AI-powered applications, and cloud-native solutions. I enjoy learning in public, contributing to open source, and using the internet and modern AI tools to continuously expand my skills. Passionate about taking ideas from concept to deployment while adapting quickly to new technologies and solving real-world problems.

Core Skills

- **Languages:** Python, HTML5, CSS3, JavaScript, SQL, Bash
- **Frameworks & Libraries:** FastAPI, Flask, SQLAlchemy, Pydantic
- **AI & Machine Learning:** Generative AI, LLM Integration, Prompt Engineering, TensorFlow, Scikit-learn, Pandas, NumPy, FAISS, ChromaDB, RAG Systems
- **Identity & Security:** Keycloak, OAuth2, OpenID Connect (OIDC), JWT, RBAC, SSO
- **Databases & Storage:** PostgreSQL, MySQL, Redis, DynamoDB, MongoDB
- **Cloud & DevOps:** AWS (EC2, S3, IAM, VPC, CloudFront), Huawei Cloud, Docker, GitHub Actions, Infrastructure as Code (Terraform & Ansible)
- **Messaging & APIs:** REST APIs, Kafka, Webhooks, OpenAPI/Swagger
- **Testing:** Unit Testing, Integration Testing, System Testing, Pytest, Unittest
- **Tools & Operating Systems:** Linux, Git, Bash Scripting, Postman

Professional Experience

Beautyverse | Software Engineer (Dec 2025 – Present)

A backend engineer architecting and developing high-scalability backend services for BeautyVerse's business management platform and marketplace. Responsible for designing robust database structures, creating RESTful APIs with FastAPI, and implementing Keycloak for identity management, SSO, RBAC, and OAuth2/OpenID Connect authentication. Integrate payment gateways and external services while automating deployments utilizing AWS, Docker, Linux, CI/CD pipelines, and Infrastructure as Code

Huawei Technologies | Transmission Data Analyst (Feb 2021 – Nov 2025)

Functioned as a Transmission Data Analyst, where I architected and managed centralized pipelines and automation frameworks for large-scale telecom infrastructure. I engineered cloud-integrated reporting and analytics tools on Huawei Cloud, automating workflows to enhance data reliability and auditability while utilizing Linux for administration and scripting. I leveraged Docker to containerize analytics services for efficient deployment, applying data-driven strategies to maximize network efficiency, delivery speed, and cost-effectiveness.

Education - University of the Witwatersrand

- MSc Computational & Applied Mathematics (2021–2024)
- BSc (Hons) Mathematical Science (2019)
- BSc Computational & Applied Mathematics (2016–2018)

Professional Certifications

- AWS:
 - Solutions Architect – Associate
 - Machine Learning Engineer – Associate
 - AI Practitioner
 - Generative AI Developer – Professional (Early Adopter)
- Linux Foundation: LFEL1004 – Authentication & Authorization for Web/API
- SAS: Certified Associate – Programming Fundamentals Using SAS 9.4

SoftwareVerse | Open source | Development

Contributed production-ready Python code to open-source projects within the SoftwareVerse ecosystem, including **Userverse**, a multi-tenant identity and organization management platform. Designed GitHub Actions CI/CD workflows and currently develop an Infrastructure-as-Code (IaC) solution to automate secure, repeatable AWS deployments, with support planned for additional cloud providers.

MedDesk | AI Clinical Intake Platform | Project

Built an AI-powered clinical intake platform that enables patients to report symptoms through conversational AI before consultations. Developed FastAPI backend services to generate structured SOAP notes, identify clinical red flags, and streamline practitioner workflows.

LetsGo South Africa | AI-Powered Tourism Platform | Project

Developed a full-stack tourism platform enabling businesses to manage travel packages while customers browse destinations, submit enquiries, and interact with an AI travel assistant. Built cloud-ready backend services with FastAPI, PostgreSQL, authentication, media storage, and deployment infrastructure.